Biospectra : ISSN: 0973-7057, Vol. 18(2), September, 2023, pp. 143-152 **Animal Science**
An International Biannual Refereed Journal of Life Sciences

ISSN : 0973-7057

Ecological risk assessment of selected heavy metals in the fish inhabitants at Raj Ghat, Varanasi, Uttar Pradesh, India

Shreya & G. B. Chand*
Aquatic Toxicology Laboratory, Department of Zoology, Patna University, Patna, Bihar, India

Received : 25th July, 2023 ; Revised : 25th August, 2023
DOI:-https://doi.org/10.5281/zenodo.12525008

Abstract- Varanasi city is considered as the spiritual capital of India. It draws the attention of Hindu pilgrims to take the holy bathe in Ganges River sacred water and perform funeral rights. Raj Ghat being the northernmost Ghat of Varanasi has recently been renovated by the state government and Varanasi municipal corporation for tourism and religious rituals. The exposure of this Ghat with heavy metals was reported to be minimal in contrast to other neighboring Ghats and Ganges River water near Varanasi. A few literatures also reported the presence of few heavy metals in the water bodies of the Ghats of Varanasi. The present study was carried out to access the bioaccumulation of selected heavy metals *viz.* mercury, lead and arsenic in the hepatic tissues of three predominant fish inhabitants *Labeo rohita* (order-Cypriniformes), *Clarias batrachus* (order- Siluriformes), *Tilapia mossambica* (order-Perciformes), as well as water and soil samples at this site. Fishes (n=18) were procured from the selected site at Raj Ghat, brought to the Central Research laboratory, Patna Women's College, Patna, Bihar and stored at -20ºC for further AAS analysis. The soil and water samples were also collected and processed for flame atomic absorption spectrophotometry. The soil sample showed the presence of mercury (0.01±0.001mg/l), lead (0.002±0.003mg/l) and arsenic (0.013±0.003mg/l). The water sample showed the presence of mercury (0.04±0.0023 mg/l), lead (0.067±0.01mg/l) and arsenic (0.010±0.001 mg/l). Among the fish samples maximum accumulation of mercury was reported in the muscles of *C. batrachus* followed by *L. rohita.* Maximum accumulation of lead was reported in muscles of *C. batrachus* and least in *T. mossambica.* The maximum accumulation of arsenic was reported in muscles of *L. rohita* followed by *C. batrachus* and *T. mossambica* respectively. Hence, the findings of the present study reveal a higher bioaccumulation of heavy metals in the fish muscles than the permissible limit. It envisages about regular and systematic bio-monitoring and strict law implementation along with scientific awareness of the local people for mitigating heavy metals pollution load at Raj Ghat, Varanasi, U.P., India.

Key words: Ecological, Risk assessment, Heavy metals, Ganga River, Raj Ghat, Varanasi, Bioaccumulation.

INTRODUCTION

The pollution of natural aquatic resources with heavy metals released from domestic, industrial and other anthropogenic activities has become a major problem over the past few decades.[1] Hazardous effects of heavy metals on aquatic organisms can be detected by performing several toxicity tests that help in establishing a dose-response relationship[2-4] and predicting acute damage to aquatic fauna and regulating toxic chemical discharges into the water bodies.[5] Due to detrimental effects of metals on aquatic organisms, it is very much important to monitor their bioaccumulation patterns. This will give an indication of temporal and spatial extent of metal accumulation, and their potential impact on health of the humans.[6]

Fishes are the top consumer in the aquatic food chain and can accumulate a large amount of heavy metals in their

*Corresponding author :
Phone : 9431406660
E-mail : gbchand@pup.ac.in

body.[7] These chemicals when absorbed are transported by the blood to either a storage point (bones and liver) or further transported to other organs like kidney, gills, fat *etc.*[8]

'Varanasi' is one of the oldest continually inhabited cities in the world.[9] Since ancient times the city has been an important centre for Hindu devotion, pilgrimage, mysticism and philosophy contributing to its cultural importance. The city is known worldwide for its many Ghats, steps leading down the steep river bank to the water where pilgrims perform rituals. Importantly the Ghats like Dashashwamedh Ghat, the Panchganga Ghat, the Manikarnika Ghat, and the Harishchandra Ghat, have a great mythological significance for the Hindus. Varanasi is located at an elevation of 80.71 metres (264.8 ft.) in the centre of the Ganges valley of North India, in the Eastern part of the state of Uttar Pradesh, along the left crescent-shaped bank of the Ganges, averaging between 15 metres (50 ft.) and 21 metres (70 ft.) above the river.[10] Located in the Indo-Gangetic Plains of North India, the land is very fertile because low-level floods in the Ganges continually replenish the soil.[11] Varanasi is situated between the Ganges confluences with two rivers- the Varuna and the Assi stream. The distance between the two confluences is around 2 miles (4 km), and serves as a sacred journeying route for Hindus, which culminates with a visit to a Sakshi Vinayak Temple.[12]

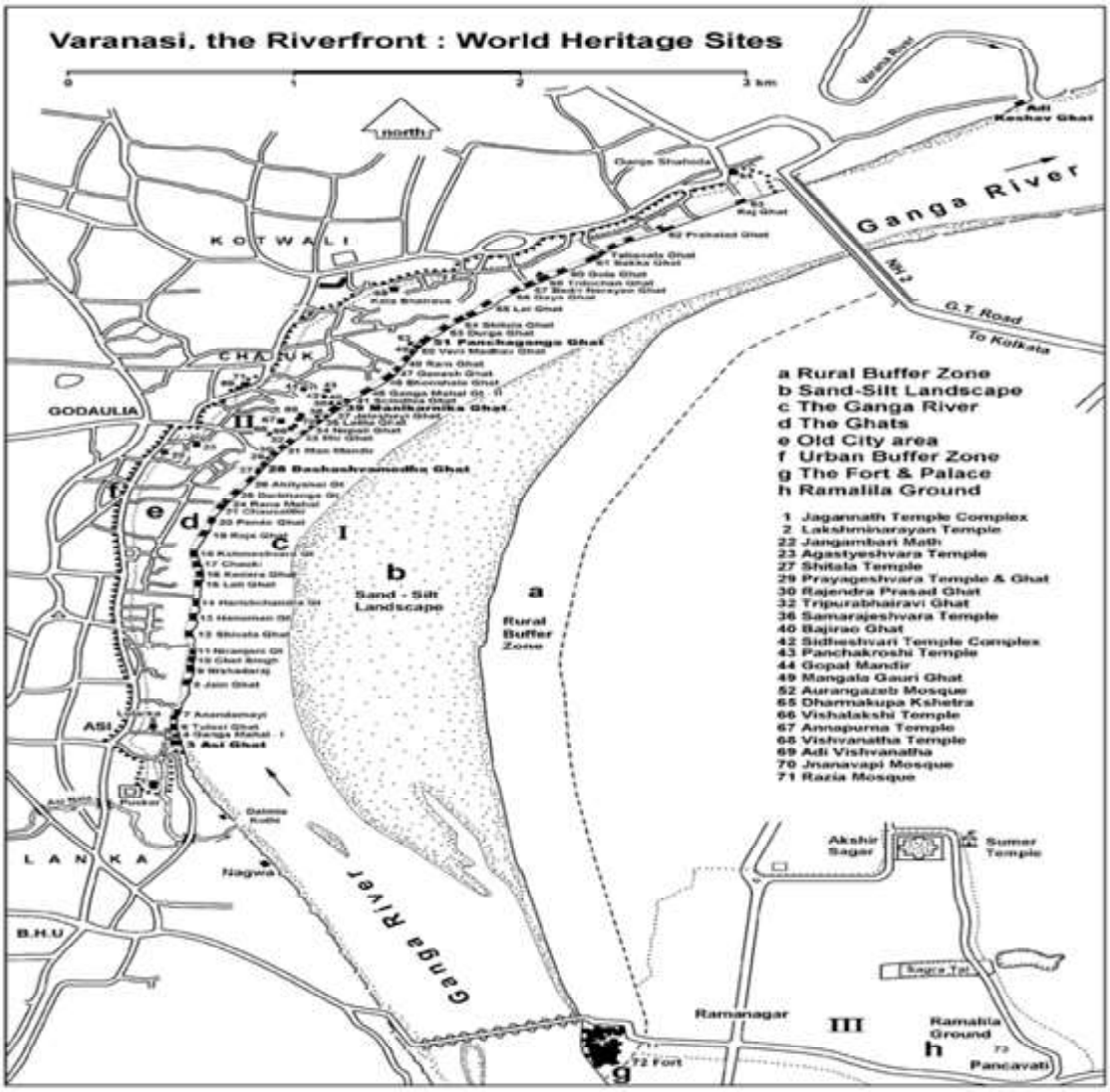


Figure 1. Map showing River Ganga (source: Ganga. The third pole) from where the Indo-Gangetic plains (IIT Kanpur) originates then the major tributaries of Ganga River (article by Mohammad Ayaz Alam) the first map of Banaras/Varanasi by Prinsep (1822) and the riverfront heritage zone and the Ganga Ghats of Varanasi (Singh Rana P.B. 1993).

Review of literature reveals the presence of arsenic (As), mercury (Hg) and lead (Pb) in the water bodies near Varanasi, Uttar Pradesh. Among the metallic pollutants, arsenic (As) is one of the most hazardous single substance toxicants in the environment, which are known to be carcinogenic and genotoxic.[13] Arsenic, in the form of trioxide, is commonly used in pharmaceuticals, pesticides, decolorizing agents and veterinary products making human beings more accessible to its exposure.[14] Owing to its ability to affect the distribution and retention of other heavy metals, mercury is one of the most hazardous heavy metals.[15] Relatively high solubility and stability of certain mercury salts in water makes them to be readily taken up and bio-transformed to methyl-mercury by certain fish species and these forms are readily absorbed through the gastrointestinal tract and become a major source of mercury exposure in humans.[16]

Lead poisoning of a person may result in abdominal pain, constipation, tiredness, headache, irritability, and loss of appetite, memory loss, pain or tingling in the hands or feet and weakness.[17] Exposure to high levels of lead may cause anemia, weakness, kidney and brain damage and a very high exposure can cause death.

Raj Ghat is the northernmost Ghats of Varanasi and is situated just before the Malviya Bridge. It is one of the most famous Ghats of Varanasi. Review of literature suggested the presence of few heavy metals in the Ganges River water along the Ghats of Varanasi, U.P. It clearly suggests that a close monitoring of even these restricted areas of the river is required periodically. In the present study, ecological risk assessment of few heavy metals has been done in soil, water and muscles of few selected fish inhabitants of Rajghat, Varanasi, Uttar Pradesh, India. It has been further correlated with the physico-chemical condition of the water. The paper also suggests some probable conservational strategies.

MATERIALS & METHODS

The Ganga River spreads between 25º 17.350' to 25º 19.678' North and 83º 00.340' to 83º 02.374' East, and covers an area of 374ha (spread between 0.5 km to 1.8 km wide, and 6.4 km long).

Varanasi is situated on the older floodplain of river Ganga which is demonstrated by a number of ponds formed by the abandonment of palaechannels. There are around 84 Ghats in Varanasi, Uttar Pradesh, India. Dashashwa medh Ghat, the Panchganga Ghat, the Manikarnika Ghat, and the Harishchandra Ghat are mythologically important.

Source:- https://www.maps.google.com



Fig. 2. Location map of study area.



Fig. 3. GPS location of study area.

Collection of water samples: Water sample was collected from the River Ganga at Rajghat, Varanasi, Uttar Pradesh. Samples were collected in good quality screw capped high density pre sterilized polypropylene bottles of 1 lit capacity, labeled properly and analyzed in laboratory for physico-chemical properties. For assessment of the water quality, monitoring was done during June 04, 2023 to June 15, 2023. The pH, temperature, total alkalinity, hardness and dissolved oxygen content of water were assessed as per the standard protocol[5]. High pure (Anal R grade) chemicals and double distilled water were used for preparing solutions for analysis of heavy metals by Atomic Absorption Spectrophotometer. Water was mixed with concentrated acids with aqua regia, 50 ml of HCl and 50 ml of HNO_3 in the ratio 1:1. The sample was oven dried by covering it with watch glass. The solution was filtered using filter paper in 100 ml volumetric flask and was diluted with distilled water up to the 100 ml mark. Now, the sample was ready to be analyzed under AAS for lead, mercury and Arsenic.

Collection and processing of soil samples: The soil sample were collected at 15cm depth around the collection site at Rajghat, Varanasi, thoroughly mixed and transferred into clean and labeled polypropylene bottles for further

analysis. The samples were mixed, gently homogenized and sieved through 2-mm-mesh sieve. The samples were first air dried, then placed in electric oven at a temperature of 40°C for about 30 minutes. The resulting fine powder was kept at room temperature for digestion.

1g of the soil sample was taken and digested in concentrated acids with aqua regia (HCl: HNO_3 = 1:1). Then the sample was oven dried by covering it with the watch glass until it became homogeneous. The resulting solution was now filtered through filter paper in 100 ml volumetric flask and diluted by distilled water till the 100 ml mark. Afterwards sample was analyzed under AAS for detection and quantification of lead, mercury and arsenic.

Collection and processing of fish samples: A total of 18 fish samples were collected from Rajghat, Varanasi, Uttar Pradesh, which included six samples each for three species namely- *Labeo rohita* (order- Cypriniformes), *Clarias batrachus* (order- Siluriformes), *Tilapia mossambica* (order- Perciformes). Average fish length was determined as 15.25 ± 1.06 for *Labeo rohita*, 22.41±0.83 for *Clarias batrachus* and 29.87±0.83 for *Tilapia mossambica*. Samples were collected in ice box and brought to Patna, Bihar for further analysis at Central Research Laboratory, Patna Women's College, Patna, Bihar.

All the samples were frozen at -20ºC until the analysis has been done. 1gm of the muscular tissue was cut near the abdomen of the different species of fish and digested in concentrated acids with aqua regia (HCl: HNO_3 = 1:1). Then the sample was covered with the watch glass and oven dried until the sample became homogeneous. The resulting solution was now filtered using filter paper in a 100 ml volumetric flask and was diluted using distilled water up to the 100 ml mark. Sample was analyzed under AAS for lead, mercury and Arsenic.

Statistical analysis: The data obtained were subjected to standard statistical analyses. Standard deviation and standard error of mean of six replicates (n= 6) were calculated. In each case data have been represented as mean ± SEM. One-way ANOVA test was done to determine the variation amongst means of different variables as well as between different individuals. The data obtained at $p<0.05$, $p<0.01$ and $p<0.001$ were found significant.

RESULTS & DISCUSSION

Fish have the ability to bio-accumulate the heavy metals, trace elements, persistent organic pollutants and residues as well as polychlorinated biphenyls (PCBs) in their tissues. The consumption of such fishes with the accumulated heavy metals is harmful to the aquatic ecosystem and humans. For the estimation of the heavy metal pollution in fresh water systems, the fish samples are considered as one of the most indicative factors. A study related to short term effects of different concentrations of two pesticides cypermethrin (CYP) and chlorpyrifos (CPF) in common carp *Cyprinus carpio* (Linnaeus, 1758) under laboratory condition have been reported.[18]

Raj Ghat is the most famous northernmost Ghats of Varanasi and is situated just before the Malviya Bridge. In 1970, the municipal corporation of Varanasi built a Pucca Ghat here to attract tourists. Many travel companies organize walks here and use the riverfront as the starting point for the boat tour. It is considered as centre of attraction for the scholars of history. Raj Ghat is home to one of the newest excavation sites of Archaeological Survey of India. Just a few meters away from the Ghat, are remains of ancient settlements dating back to 8th century BC to 18 century AD. It clearly suggests that Varanasi is much older than what it was thought to be. In medieval age, Raj Ghat used to be the residence of the kings, which justifies its name. The upper portion of the Ghat was Gahadavala fort and the lower part was used as a ferry point. Even today, one can see the dismantled fort of Rajghat which is divided by the Grand Trunk Road. The inscription of the Mauryan and Gahadavala designed Raj Ghat as one of the most sacred spots in Varanasi. It was the busiest Ghat till 12th century, but lost its importance as people shifted to the southern parts and other Ghats became more prominent. It continued to be emerged as a ferry point till the construction of the Malviya Bridge. Today, it is flooded by locals, Ghat merchants, fishermen, and boats men, who live nearby. Very recently a small pucca stage was constructed here to perform daily Aarti. The Aarti is organized by locals and follows the same process as followed by the other Ghats. The fish species mainly found in river Ganga at Varanasi, Uttar Pradesh are the Indian major carps, other carps, mullets, clupeids, feather backs, large/ small catfishes and miscellaneous fishes. Some exotic fishes are also found in the river Ganga at Varanasi like *Tilapia mossambica*.

Physico-chemical analysis of water collected from six different sites at Rajghat, Varanasi, Uttar Pradesh

Sample No.	Temperature (°C)	pH	Dissoved Oxygen (mg/l)	Total alkalinity (mg/l)	Hardness (mg/l)
1	29.1±1.026	7.70±0.679	5.226±0.659	124.0±0.1	160.0±0.3
2	27.8±2.016	7.71±0.241	4.924±1.023	122.0±0.3	162.0±0.2
3	28.3±1.059	6.69±0.358	5.226±1.2035	128.0±0.1	166.0±0.2
4	29.4±1.231	7.70±0.959	5.537±0.998	124.0±0.2	160.0±0.1
5	30.2±1.526	6.69±0.586	5.679±0.869	126.0±0.2	164.0±0.3
6	30.3±1.462	7.70±0.498	5.226±1.2035	120.0±0.3	168.0±0.1

The mean temperature of the water sample was recorded as 29.1± 1.23 (Mean ± SEM). The pH of different water samples was determined by standard electrical pH meter. A difference in pH of water sample was analyzed. The average water pH of the water sample was recorded as 7.70 ± 0.498 (Mean ± SEM). The dissolved oxygen content of the six water samples at different sites at Raj Ghat, Varanasi, (U.P.) was determined by volumetric titration methods (Winkler's method). The dissolved oxygen of the water samples recorded at six different sites at Raj Ghat, Varanasi, was found as 785 ± 0.659 mg/l; 4.924 ± 1.023mg/l; 5.226 ± 1.2035mg/l; 5.537 ± 0.998mg/l; 5.679 ± 0.869 mg/l and 5.226 ±1.2035 mg/l. The average content of dissolved oxygen was recorded as 5.226 mg/l ± 1.2035. Total alkalinity was recorded by titration method using methyl orange and phenolphthalein as two indicators. The sample was titrated with 0.02N H_2SO_4 until the colour disappears. The average phenolphthalein alkalinity was recorded as 6.0 mg/l and total alkalinity was recorded as 124 mg/l. Total hardness of water was measured by titration method using EDTA as a titrating agent and Eriochrome Black T as indicator. The total hardness of Ganga water sample near Raj Ghat was measured as 160.0 mg/l (Table-1).

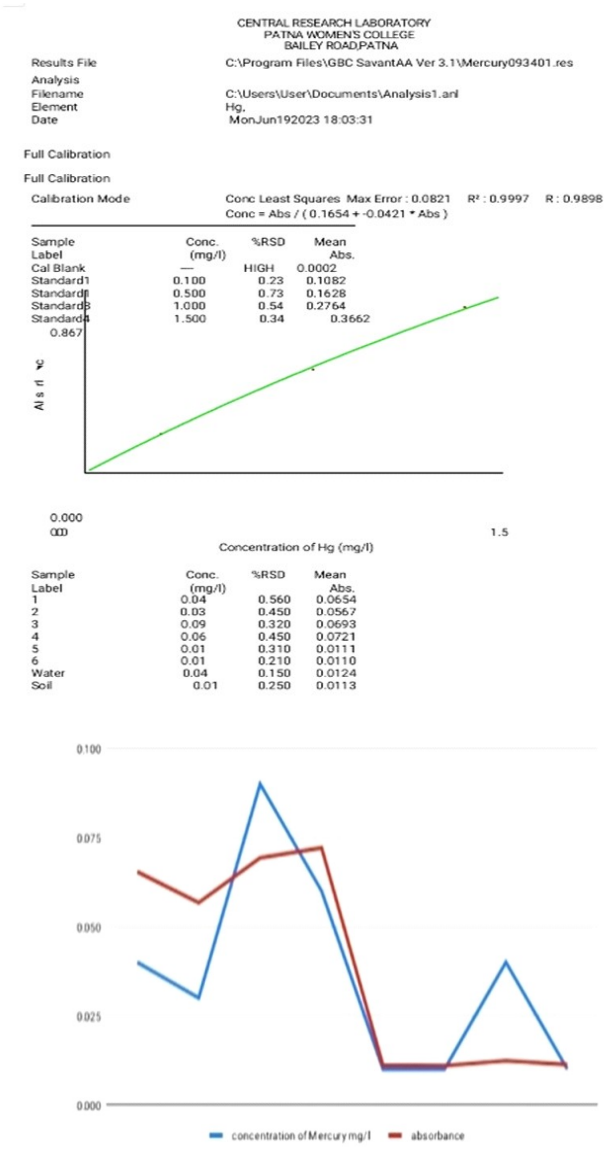
For the heavy metal assessment in water, soil and fish samples at Raj Ghat, Varanasi three heavy metals *i.e.* mercury, lead and arsenic have been considered based on the review of literature. Three fish samples *Labeo rohita*, *Clarias batrachus* and *Tilapia mossambica* were collected and their muscles were analyzed for accumulation of heavy metals.

Mercury, the most toxic heavy metals found in the environment, ranked 3rd in the list of hazardous substances of the environment after lead and arsenic by United State Environmental Protection Agency (EPA) and The Agency for Toxic Substances and Disease Registry (ATSDR). Mercury can enter into the fish body by foods through the alimentary canal, skin and gills. Even at sub-lethal concentration, it can cause structural, physiological and biochemical alterations in the physiology of nervous system. Due to lipophilic nature, it can cross blood- brain barrier and cause the nervous tissues damage. Mercury has high affinity for proteins therefore more than 90% of total mercury accumulates in the fish muscles.[19]

Mercury exert its toxic effects by interfering and displacing iron and copper from the active site of enzymes involved in energy production like mitochondrial dysfunction and oxidative damage. Mercury can also directly accelerate the oxidative destruction of cell membranes and low density lipid (LDL) cholesterol particle, at the same times, it binds and inactivates the cellular antioxidants like N-acetyl cysteine, alpha-lipoic acid, and glutathione. Because of its effects on cellular defense mechanism and energy generation, mercury may cause widespread toxicity and symptoms in different organ systems; nervous system like personality changes, tremors, memory deficits, loss of co-ordination; cardiovascular system *i.e.,* increased risk of arterial absorption, hypertension, stroke, and atherosclerosis, heart attacks, and increased inflammation; gastrointestinal tract like nausea, diarrhea, ulceration, and kidney failure.[20] Mercury also

accumulates in the thyroid and increases the risk of autoimmune disorders and may cause contact dermatitis.[21]

In the present study, the level of mercury in the water of River Ganga at Rajghat, Varanasi was found to be 0.04 ± 0.0023 mg/l. The permissible limit of mercury in drinking water has been vividly suggested by different agencies as- 0.001mg/l (World Health Organization); 0.002 mg/l (United States Environmental Protection Agency); 0.001mg/l (Indian Standard Institution & Indian Council of Medical Research) The level of mercury in the soil sample, was observed as 0.01±0.001 mg/l (Text Graph-1a).

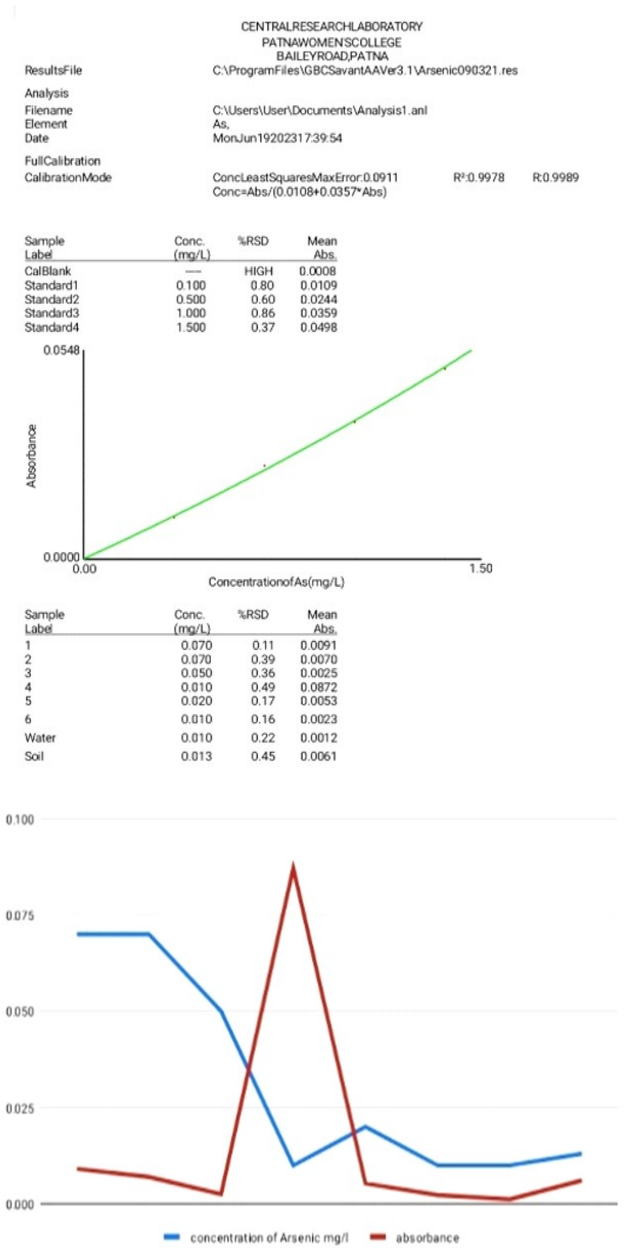


Text Graph 1a & b: Showing accumulation of Hg in different fish species viz. *Labeo rohita, Clarias batrachus* & *Tilapia mossambica,* water & soil samples.

Among the fish samples, the minimum concentration of mercury was observed in the muscles of *Labeo rohita* as 0.04±0.003 mg/l and 0.03±0.01 mg/l respectively. Our findings are in agreement of Svobodová *et al.* (1993)[22] who reported 0.002 mg/l. concentration of the inorganic form of mercury in cyprinids. In freshwater air breathing fish *Clarias batrachus*, the concentration of mercury in the muscles was found to be 0.09 and 0.06 mg/l. Similar, kind of result along with acute damage and necrosis of nephric tubules in *Clarias batrachus* due to chronic exposure of mercurial compounds have been reported[23-24]. Further increase of serum cholesterol and cortisol, aspartate aminotransferase, alanine aminotransferase, alkaline phosphatase, urea and creatinine levels and consistent decline in haemoglobin and haematocrit value in *Clarias gariepinus* after exposure to mercury have been reported.[25] In the present study, the concentration of mercury in muscles of *Tilapia mossambica* was found to be 0.01mg/l which shows its consistency with that of soil (Text graph-1b).

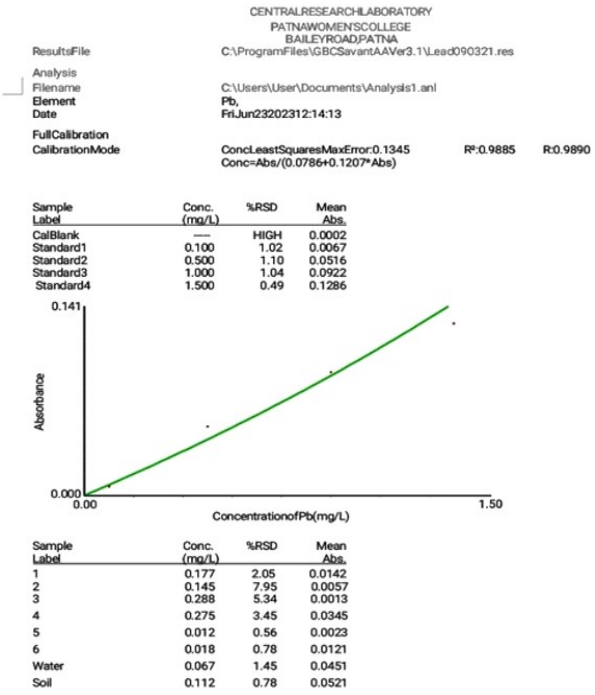
Lead is considered as hazardous heavy metal which is naturally present in the environment in combination with other elements like PbS, $PbCO_3$, $PbSO_4$. Different anthropogenic sources like metal mining, combustion of coal and gasoline, battery manufacturing, lead – arsenite pesticides, pigments in food can add lead to the environment.[26] Lead discharge from various industries, agricultural fields, street runoff, leaching of battery and municipal wastewater cause toxicity to the aquatic life.[27] The solubility of lead in water depends upon pH, salinity, hardness. Marked behavioural changes, impotency and growth retardation of the fish after sub-lethal lead exposure were marked.[28] Low concentration of lead nitrate in *Clarias gariepinus* showed histological distortion of gill and liver tissues.[29-31] Prolonged exposure showed a remarkable alteration in cholesterol and lipid content of liver, brain and gonads in *Clarias batrachus.*[32] A decline in the RBC count and hematocrit value have been reported in Nile tilapia (*Oreochromis niloticus*). Besides an oxidative stress causing synaptic damage and neurotransmission malfunctions as well as alteration in immunological parameter in *Tinca tinca* have also been recorded.[33-34] Lead bioaccumulation in fish mainly occurs in the liver, spleen, kidney and gills.[35] In the present study, lead concentration in the water and soil sample was recorded as 0.067±0.010 mg/l and 0.112±0.003 mg/l respectively (Text graph-2a).

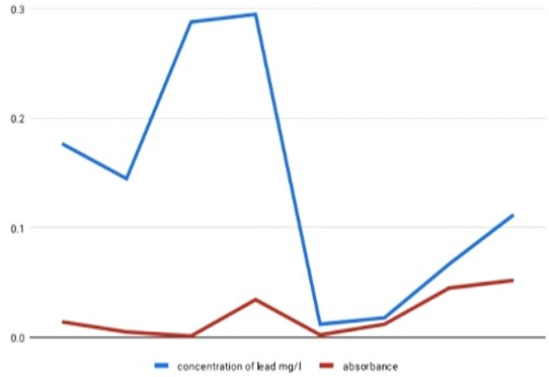
Among the fish samples, muscles of *Labeo rohita* showed 0.177±0.002 mg/l and 0.145±0.001 mg/l of lead accumulation. In the freshwater air breathing fish *Clarias batrachus,* the lead concentration was found to be 0.288±0.001mg/l and 0.275±0.0012 mg/l while in *Tilapia mossambica*, it was recorded as 0.012±0.001mg/l and 0.067±0.001mg/l respectively (Text graph 2b).



Text Graph 2a & b: showing accumulation of Pb in different fish species viz. *Labeo rohita, Clarias batrachus* & *Tilapia mossambica*, water & soil samples.

Arsenic is ubiquitous metalloid; it is being released in the aquatic environment from various anthropogenic sources like manufacturing companies, smelting operations, power plants etc. The agricultural run-off is one of the major sources. Continuous exposure of freshwater fish to the low concentration of arsenic resulted in bioaccumulation in the liver and kidney tissues.[36] In *Oreochromis mossambicus,* arsenic exposure showed histo-pathological alterations in gills as epithelial hyperplasia, lamellar fusion, epithelial lifting edema, desquamation and necrosis while in liver tissues macrophage infiltration, vascularization, hepatocyte shrinkage, dilation of sinusoids, vascular degeneration, nuclear hypotrophy and focal necrosis were prominently recognized.[37-39] Acute exposure of sodium arsenite to *Clarias batrachus* showed disturbance in hematopoiesis, disruption of the erythrocyte membrane, impaired iron uptake by erythrocytes and hemolysis.[40] Arsenic induced alteration in T-cells and B-cells mediated functioning and diminished phagocytic functioning have been reported in the cat fish.[41] Arsenic content in the water and soil samples at Raj Ghat, Varanasi, was found to be 0.010±0.001 mg/l. and 0.013±0.003 mg/l respectively. In the muscles of *Labeo rohita, Clarias batrachus*, and *Tilapia mossambicus* the concentration of arsenic was found to be 0.070±0.002 mg/l, 0.050±0.02 mg/l and 0.020± 0.001 mg/l respectively (Text graph 3a & b).





Text Graph 3a & b: showing accumulation of As_2O_3 in different fish species viz. *Labeo rohita, Clarias batrachus* & *Tilapia mossambica*, water & soil samples.

Strategies to improve the heavy metal pollution of River Ganga- The heavy metal pollution of the River Ganga has drawn the attention of the scientists and others concerned with the issues related to environmental degradation. It is high time for all the scientists, technocrats, River conservationist, politicians, Government and all the other stakeholders to unite and strategize for the improvement in the pollution status of River Ganga with particular reference to heavy metal pollution.

Some of the exclusive strategies to improvise the health status of River Ganga with particular reference to Raj Ghat can be summarized as:

1. Regulatory standards for emission and discharges from different industries should be strict in adjoining area.

2. Recycling of wastewater containing heavy metals needs to be given greater importance not only from environmental and health concerns but also as a resource conservation initiative.

3. Monitoring of wastewater from toxic heavy metal processing units of the different industries needs to be executed more vigorously.

4. The government should outline a plan or strategy to comprehensively survey the Ganga in order to identify and specify the sources of the pollution.

5. The effect of heavy metals present in water and sediments of river Ganga on the biotic community were studied only to a limited extent. More elaborative study should be promoted through grant of major funding.

6. The occurrence of very large amount of heavy metal pollutants into surface water and sediment can affect the self-purifying nature of the river. As soon as the river loses its self-purifying nature, the growth of high level of pathogenic bacteria results. The area needs to be focused separately.

7. The regular and systematic monitoring and strict law implementation are required to achieve sustainable way for enhancing the water quality.

8. At last, awareness should be raised among the common people through various scientific and conservational programmes of Government and NGOs.

CONCLUSION

The hazards of bioaccumulation and bio-magnification of the heavy metals make them a big risk to human health and welfare. Hence, it is necessary that steps be taken to minimize the metallurgical effluent load deposited into the river Ganga. The findings of the present study reveal a higher bioaccumulation of heavy metals in the fish muscles than the permissible limit. It envisages about regular and systematic bio-monitoring and strict law implementation along with scientific awareness of the local people for mitigating heavy metals pollution load at Raj Ghat, Varanasi, U.P., India.

ACKNOWLEDGEMENT

The authors are thankful to Head, Department of Zoology, Patna University for providing infra structural facilities and the Principal, Patna Women's college for giving access to CRL, Patna Women's College, Patna, Bihar.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

REFERENCES

1. **Waqar K., Ahmad I., Kausar R., Tabassum T. and Muhammad A. 2013.** Use of bio-remediated sewage effluent for fish survival. *Int J Agric Biol,* **15**: 988 992.
2. **Akter M. S., Ahmed K., Akhand A. A. and Islam M. 2008**. Acute toxicity of Arsenic and mercury to fresh water climbing perch, *Anabas testudineus (Bloch)*. *World J Zool*, **3:**13-18.
3. **Aktar M. W., Paramasivam M., Ganguly M., Purkait S. and Sengupta D. 2010.** Assessment and occurrence of various heavy metals in surface water of Ganga river around Kolkata: A study for toxicity and ecological impact. *Environ. Monit. Assess.* **160:** 207e213.

4. **Javed M. and Usmani N. 2019.** An Overview of the Adverse Effects of Heavy Metal Contamination on Fish Health DOI: 10.1007/s00244-006-0237-6 *Proceedings of the National Academy of Sciences, India - Section B: Biological Sciences* **89((2):**389–403.

5. **APHA 2023.** Standard Methods for the Examination of Water and Wastewater, 24th Edition. Edited by William C. Lipps, Ellen Burton Braun-Howland and Terry E. Baxter.

6. **Ladipo M.K., Ajibola O.V. and Oniye S.J. 2012.** Spatiotemporal assessment of metal concentration in fish and periwinkles in selected locations of lagos lagoon, Nigrria. *J Environ Chem Ecotoxicol,* **4:**161-169.

7. **Chezhian N., Kabilan T., Kumar T.S. and Senthamilselvan D. 2010.** Impact of Chemical Factory Effluent on the Structural Changes in Gills of Estuariane Fish, *Mugilcephalus. World Appli. Sci. J.* **9:** 922-927.

8. **Dural M., Goksu M. Z. L. and Ozak A. A. 2007.** Investigation of heavy metal level In economically important fish species captured from the Tuzla Lagoon. *Food Chem,* **102:** 415-421.

9. **Chirico S. and Kerry P. C. 2012.** "Banaras", in Juergensmeyer, Mark; Roof, Wade Clark (eds.), *Encyclopedia of Global Religion,* **1:** 114-116, ISBN 9780761927297.

10. **Mohanty B. 1993.** Urbanisation in Developing Countries: Basic Services and Community Participation. *Concept Publishing Company*. ISBN 978-81-7022-475-4. Archived from the original on 4 April 2023. Retrieved 6 March 2016.

11. **"Indo-Gangetic Plain". Encyclopaedia Britannica. 2018.** Archived from the original on 26 December 2018. Retrieved 26 December 2018.

12. **Kishore K. 2008.** Holy Ganga. Rupa Publications. ISBN 978-81-291-3328-1. Archived from the original on 4 April 2023. Retrieved 6 March 2016.

13. **Chen Y.C., Su H. J., Guo Y. L., Hsueh Y., Smith T. J., Ryan L. M., Lee M. and Christiani D. C. 2003.** Arsenic methylation and bladder cancer risk in Taiwan. doi: 10.1023/a:1023905900171. *Cancer Causes Control*. **14(4):**303-10.

14. **Bohrer D., Becker E., Nascimento P.C., Mörschbächer V., Machado de Carvalho L. and Marques da Silva M. 2006.** Arsenic release from glass containers by action of intravenous nutrition formulation constituents. *International Journal of Pharmaceutics* **315 (1–2):** 24-29.

15. **ATSDR 2007.** Toxicological Profile for Lead. *Goergia: Agency for Toxic Substance and Disease Registry,* US Department of Health and Humans Services; Available online: https://ec.europa.eu/oceans-and-fisheries/facts-and-figures/facts-and-figures-common-fisheries_policy/Consumption_en (accessed on 15 December 2021).

16. **Peterson A. S., David V. P. Van Sickle J. Hughes R. M. 2007.** Mercury concentration in Frozen whole fish homogenates in insensitive to holding time. *Arch Environ Contam Toxicol* doi: 10.1007/s00244-006-0237-6. **53(3):**411-7.

17. **ATSDR 2005.** Lead toxicity. *Georgia: Agency for Toxic Substances and Disease Registry*, U.S. Department of Health and Human Services.

18. **Georgieva E., Yancheva V., Stoyanova S., Velcheva I., Iliev I., Vasileva T., Bivolarski V., Petkova E., László B., Nyeste K, *et al.* 2021.** Which Is More Toxic? Evaluation of the Short-Term Toxic Effects of Chlorpyrifos and Cypermethrin on Selected Biomarkers in Common Carp (*Cyprinus carpio,* Linnaeus 1758). *Toxics*, **9:** 125.

19. **Bradley M. A., Barst B. D. and Basu N. 2017.** A Review of Mercury Bioavailability in Humans and Fish. *Int J Environ Res Public Health*. doi: 10.3390/ijerph14020169. **14(2):** 169.

20. **Galanis A., Karapetsas A. and Sandaltzopoulos R. 2009.** Metal-induced carcinogenesis, oxidative stress and hypoxia signaling. *Mutation Research.* **2:** 5-31.

21. **ATSDR. 2001.** Toxicological Profile for Asbestos. *Georgia: Agency for Toxic Substances and Disease Registry*, Department of Health and Human Services, Public Health Services; 2001.

22. **Svobodova Z. 1993.** Water Quality and Fish Health. FAO, Rome, EIFAC technical paper http://www.fao.org/3/t1623e/t1623e00.htm No. **54:** 67.

23. **Kirubagaran R. and Joy K. P. 1988.** Toxic effects of three mercurial compounds on survival, and histology of the kidney of the catfish *Clarias batrachus* (L.) DOI: 10.1016/0147-6513(88)90069-3 *Ecotoxicol Environ Saf.* **15(2):**171-9.

24. **Kumari K. 2022.** Assessment of Haematological and Certain Biochemical changes in Freshwater air breathing

fish *Clarias batrachus* (Linn.) at different Exposure levels of Mercury. *PhD thesis.*awarded from, Faculty of Science, Patna University, Patna, Bihar (INDIA).

25. **Monna F., Camizuli E., Revelli P., Biville C., Thomas C., Losno R., Scheifler R., Bruguier O. Baron S. . Chateau C., Ploquin A. and Alibert P. 2011.** Wild Brown Trout Affected by Historical Mining in the Cévennes, France National Park. https://doi.org/10.1021/es200755n *Env. Sci. Tecl.* **45(16):** 6823–6830.
26. **Abadin H., Ashizawa A., Stevens Y.W., Llados F., Diamond G., Sage G., Citra M., Quinones A., Bosch S. J. and Swarts S. G. 2007.** Toxicological Profile for Lead Atlanta (GA): *Agency for Toxic Substances and Disease Registry (ATSDR) Toxicological Profiles*, *U.S.*PMID: **24049859** Bookshelf ID: NBK158766.
27. **Sepe A., Ciaralli L., Ciprotti M., Giordano R., Funari E. and Costantini S. 2003.** Determination of cadmium, chromium, lead and vanadium in six fish species from the Adriatic Sea. doi: 10.1080/0265203031000069797. *Food Addit Contam.* **20(6):**543-52.
28. **Afshan S., Ali S., Shaista Ameen U., Farid M., Bharwana S. A, Hannan F. and Ahmad R. 2014.** Effect of Different Heavy Metal Pollution on Fish *Research Journal of Chemical and Environmental Sciences*. **2(1):** 74-79.
29. **Biswas S. and Ghosh A. R. 2016**. Lead induced histological alterations in ovarian tissue of freshwater teleost *Mastacembelus pancalus* (Hamilton). DOI: 10.7439/ijasr.v2i1.2797. *International Journal of Advances in Scientific Research.* **2(1):**45.
30. **Olojo E. A. A., Olurin K. B., Mbaka G. and Oluwemimo A. D. 2005.** Histopathology of the gill and liver tissues of the African catfish *Clarias gariepinus* exposed to lead. *African journal of biotechnology* **4(1):**117-122.
31. **Olojo E. A. A., Abass A. A., Olurin K. B., Mbaka G. 2012.** The potential use of certain protein metabolism parameters as biomarkers of heavy metal (lead) stress in the African catfish, *Clarias gariepinus*. *Agric. J.* **7:**316–322.
32. **Katti S. R. and Sathyanesan A. C. 1983.** Lead nitrate induced changes in lipid and cholesterol levels in the freshwater fish *Clarias batrachus.Toxicology Letters* **19(1-2):** 93-96.
33. **Lee J. W., Choi H., Hwang U. K., Kang J. C., Kang Y. J., Kim Kwang II and Kim J. H. 2019.** Toxic effects of lead exposure on bioaccumulation, oxidative stress, neurotoxicity, and immune responses in fish: A review. *Environmental Toxicology and Pharmacology.* **68:**101-108.
34. **Shah S. L. and Altindag A. 2005.** Alterations in The Immunological Parameters of Tench (*Tinca tinca L.* 1758) After Acute and Chronic Exposure To Lethal and Sublethal Treatments With Mercury, Cadmium and Lead. *Turk J Vet Anim Sci.* **29:** 1163-1168.
35. **Creti P., Trinchella F. and Scudiero R. 2010.** Heavy metal bioaccumulation and metallothionein content in tissues of the sea bream *Sparus aurata* from three different fish farming systems. DOI 10.1007/s10661-009-0948-z. *Environ Monit Assess*. **165:**321–329.
36. **Kumari B., Kumar V., Sinha A.K. and Ghosh A. 2017.** Toxicology of arsenic in fish and aquatic system. *Environmental Chemistry Letters.* DOI: 10.1007/s10311-016-0588-9 **14(55):** 1-22.
37. **Ahmed M. K., Habibullah-Al-Mamun M., Hossain M. A., Arif M., Parvin E., Akter M. S., Khan M. S. and Islam M. M. 2012.** Assessing the genotoxic potentials of arsenic in tilapia (*Oreochromis mossambicus*) using alkaline comet assay and micronucleus test. *Chemosphere* **84:**143–149.
38. **Ahmed M. K., Habibullah-Al-Mamun M., Parvin E., Akter M. S. and Khan M. S. 2013.** Arsenic induced toxicity and histopathological changes in gill and liver tissue of freshwater fish, tilapia (*Oreochromis mossambicus*). *Exp Toxicol Pathol* **65(6):**903–909.
39. **Ahmed S. A., Faruquee M. H., Khan M. H. 2020.** Twenty years of arsenic contamination and Arsenicosis patients in a village of Bangladesh. *American Journal of Public Health Research.* **8(6):** 190-196.
40. **Tripathi S., Sahu D. B., Kumar R. and Kumar A. 2003.** Effect of acute exposure of sodium arsenite (Na_2AsO_3) on some hematological parameters of *Clarias batrachus* (common Indian catfish) *in vivo*. *Indian J. Environ Health.* **45:**183–188.
41. **Ghosh D., Bhattacharya S. and Mazumder S. 2006.** Perturbations in the catfish immune responses by arsenic: organ and cell specific effects. *Comp Biochem Physiol C.* **143:**455–463.

***